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Fatih Berkay Sarpkaya

Education

2022–Present **Doctor of Philosophy (Ph.D.)**, *New York University Tandon School of Engineering*, Brooklyn, NY
Expected Graduation: May 2027
Major: Electrical & Computer Engineering | Advisor: Professor Shivendra Panwar | **GPA: 3.949/4.00**

2017-2022 **Bachelor of Science (B.Sc.)**, *Bilkent University*, Ankara, Turkey
Major: Electrical and Electronics Engineering, **CGPA: 3.77/4.00**

Research Interests

Transport Protocols & Congestion Control | Low-Latency Networking (L4S) | Internet Architecture & Traffic Engineering | Data Center & AI Networking | Network Performance Modeling & Measurement

Work & Research Experience

Summer 2026 **Microsoft Research**, *AI Network Observability Research Intern*, Redmond, WA
Performance Modeling and Analysis of AI Accelerator Infrastructure — Managers: Anqi Guo, Nicholas Lurski, Pratik Marolia

- Developed and calibrated an FPGA-based emulation framework for AI accelerator collective communication workloads.
- Modeled communication pipelines and dependencies for collective operations, validating emulation behavior against hardware measurements.
- Characterized collective communication performance across message sizes, identifying synchronization and communication overheads affecting end-to-end latency.

Summer 2025 - Fall 2025 **Nokia Bell Labs**, *Network Reliability Research Intern (Summer) / Co-op (Fall)*, Murray Hill, NJ
Congestion Control for Public High-Performance WANs — Managers: Andrea Francini, Bilgehan Erman, Gary Atkinson

- Evaluated CUBIC, BBRv1, and BBRv3 for predictable large data transfers over public high-performance WANs. [HPSR'26]
- Designed large-scale FABRIC testbed experiments using Linux and DPDK-based traffic shaping and measurement tools; modified and optimized Linux kernel congestion control modules.

Summer 2024 **InterDigital**, *Research & Innovation Wireless Intern*, New York, NY
Flexible Packet Coding Performance Analysis & Generative Semantic Communication Literature Review — Manager: Pascal Adjakple

- Implemented various network coding and packet duplication simulations in MATLAB, and analyzed performance results.
- Conducted a literature review on generative AI in semantic communication and its potential impact on modern communication systems.

Sep. 2022 – Present **NYU Wireless Research Center**, *Graduate Researcher*, Brooklyn, NY

- Transport & Low-Latency Networking: evaluation and modeling of congestion control coexistence across BBR, TCP Prague, CUBIC/Reno, AQMs, and partial L4S deployments [IFIP Networking'25] [PAM'25] [ANRW'24]
- Internet Architecture & Traffic Engineering: multi-domain low-latency service using traffic isolation, SRv6 path steering, and BGP signaling [ICNP'26]
- Reliable Communications: adversarial channel capacity and robust communication using random interleaving and thresholded LLR decoding [Asilomar'24]

Publications

Jan. 2027 **When Coupling Backfires: BBR-Induced L4S Degradation under DualPI2**
Fatih Berkay Sarpkaya, Fraida Fund, Shivendra Panwar
IEEE Consumer Communications & Networking Conference (CCNC 2027) – LATENCY Workshop

Oct. 2026 **Extending Low Latency Service Across the Internet**
Harkirat Singh*, **Fatih Berkay Sarpkaya***, Hakan Gulec, Fraida Fund, Shivendra Panwar
IEEE International Conference on Network Protocols (ICNP 2026) *Equal contribution [Link]

June 2026 **Evaluation of TCP Congestion Control for Public High-Performance Wide-Area Networks**
Fatih Berkay Sarpkaya, Andrea Francini, Bilgehan Erman, Shivendra Panwar
IEEE High Performance Switching and Routing (HPSR 2026) [Link]

May 2025 **BBR's Sharing Behavior with CUBIC and Reno**
Fatih Berkay Sarpkaya, A. Srivastava, Fraida Fund, Shivendra Panwar
IFIP Networking 2025 [Link]

March 2025 **To Adopt or Not to Adopt L4S-Compatible Congestion Control? Understanding Performance in a Partial L4S Deployment**
Fatih Berkay Sarpkaya, Fraida Fund, Shivendra Panwar
Passive and Active Measurement Conference (PAM 2025) [Link]

Oct. 2024 **Capacity of a Binary Channel with a Time-Bounded Adversary**
M. Ying, **Fatih Berkay Sarpkaya**, S. Bakirtas, Elza Erkip, T. Rappaport, Sundeep Rangan
58th Asilomar Conference on Signals, Systems, and Computers [Link]

July 2024 **To Switch or Not to Switch to TCP Prague? Incentives for Adoption in a Partial L4S Deployment**
Fatih Berkay Sarpkaya, A. Srivastava, Fraida Fund, Shivendra Panwar
Applied Networking Research Workshop (ANRW 2024) [Link]

Relevant Coursework and Tools

NYU - ECE Probability & Stochastic Processes | Internet Architecture & Protocols | Network Design and Algorithms | Data Center & Cloud Computing | Machine Learning | Network Modeling and Analysis | Digital Communication | Wireless Communication | Introduction to Wireless Networking | AI/ML for Network Traffic Analysis

Bilkent - EE Computer Networks | Telecommunications I | Telecommunications II | Probability and Statistics | Statistical Learning and Data Analytics

Tools Python | MATLAB | Linux | FABRIC Testbed | DPDK | CloudLab | GOAL

Awards & Honors

2026 **Dante Youla Award for Graduate Research Excellence**, NYU Electrical and Computer Engineering

2022-Present School of Engineering **Fellowship** at NYU Tandon School of Engineering

2017-2022 Comprehensive **Scholarship** at Bilkent University

2017 Ranked **450th** in **University Entrance Exam (LYS)** among 2 million high school students.

Teaching & Mentoring

2023–2024 **Course Assistant**, *Internet Architecture & Protocols; Communication Networks*, NYU Tandon School of Engineering

Summer 2023 **Research Mentor**, *ARISE High School Summer Research Program*, NYU Tandon School of Engineering

Additional Activities

2018-2022 Active Member of Bilkent University IEEE Student Branch

July 2019 Engaged in short-term volunteer work supporting children with disabilities through the European Voluntary Service (EVS) in Macedonia.